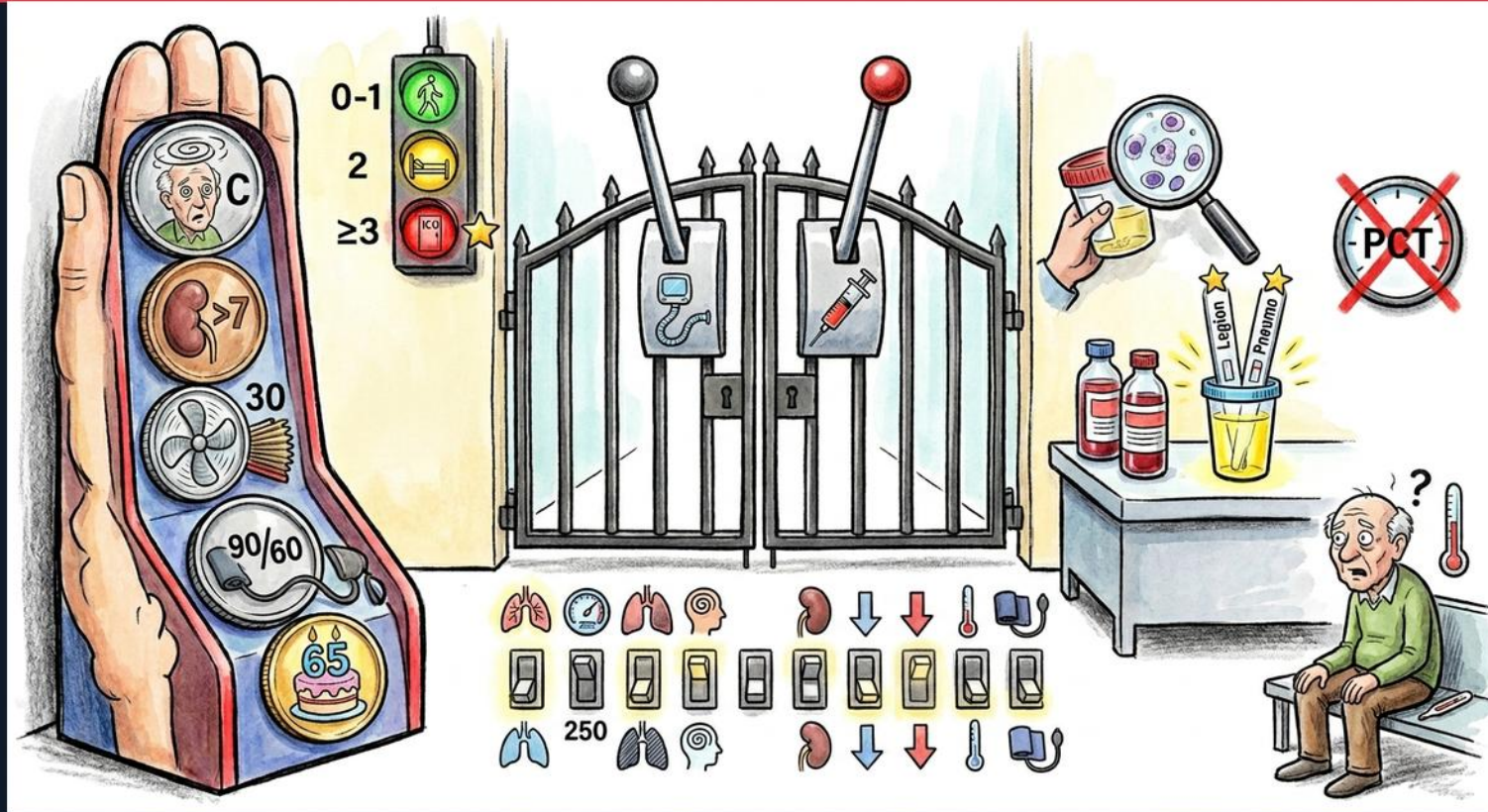
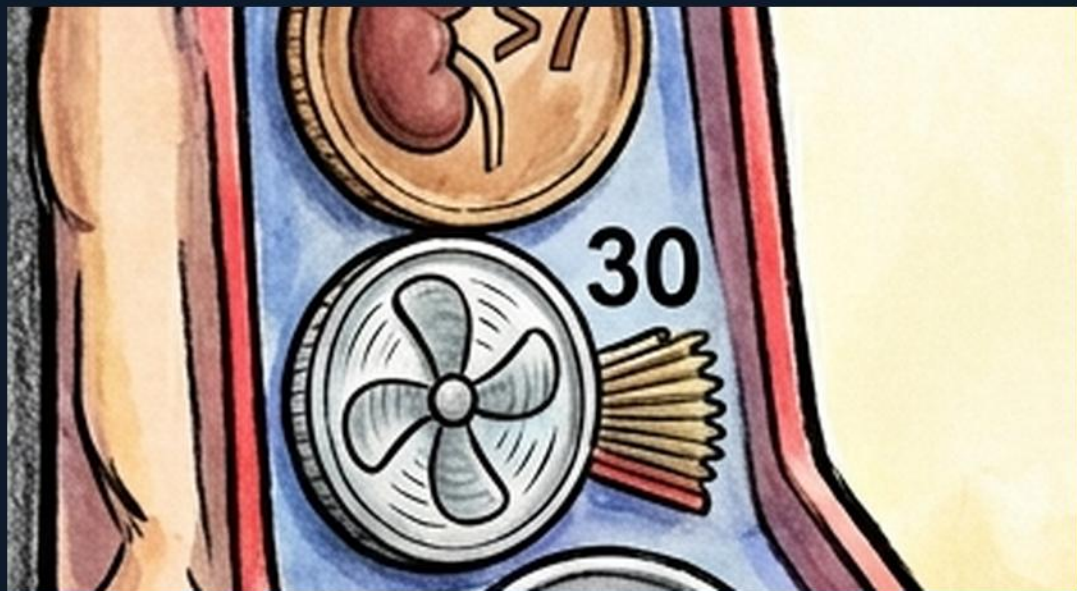


Pneumonia — Severity (CURB-65, ICU) & Diagnostics





CURB-65 hand of 5 buttons

WHAT YOU SEE

A hand-shaped arcade panel with five buttons — CURB-65: Confusion, Urea >7 , RR ≥ 30 , BP low, age ≥ 65 .

WHAT IT ENCODES

CURB-65 is a 5-point mortality/disposition tool, 1 point each for Confusion, Urea >7 mmol/L (BUN >19 mg/dL), Respiratory rate ≥ 30 , low Blood pressure (SBP <90 or DBP ≤ 60), and age ≥ 65 . It estimates 30-day mortality and is simpler than the PSI/PORT score.

THE BOARD TRAP

"choose antibiotics because CURB-65 is high" -> wrong; the score guides SITE OF CARE, not drug choice — and it can underestimate a young hypoxemic patient (no oxygenation variable).



C — Confusion button

WHAT YOU SEE

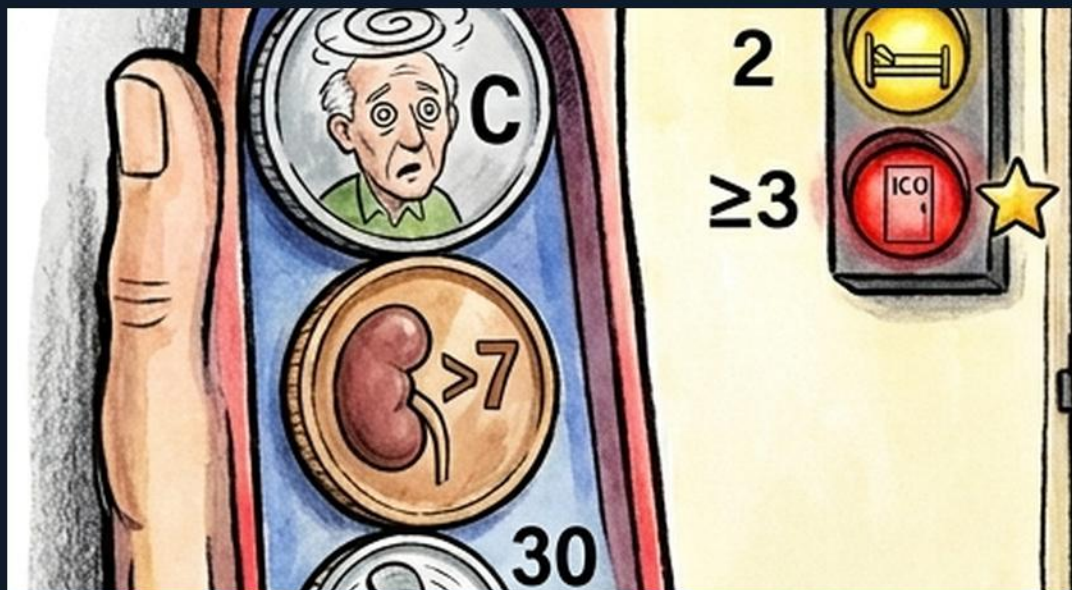
Top button: a confused elderly face marked 'C' — new-onset confusion.

WHAT IT ENCODES

New-onset confusion/disorientation scores 1 point, reflecting severe disease and/or hypoperfusion. Especially important in the elderly, in whom altered mental status may be the dominant or only presentation of pneumonia.

THE BOARD TRAP

"baseline dementia" or "just a UTI" for new confusion -> wrong; in an older adult it may be the only clue to pneumonia/sepsis. It must be NEW confusion to count.



U — Urea >7 button

WHAT YOU SEE

Second button: a kidney with '>7' — urea, not creatinine.

WHAT IT ENCODES

Blood urea >7 mmol/L (BUN >19-20 mg/dL) scores 1 point, marking end-organ hypoperfusion/dehydration and worse prognosis — one reason a metabolic panel is in the initial CAP workup.

THE BOARD TRAP

"score the creatinine" -> wrong; it is UREA/BUN — a normal creatinine with elevated BUN still earns the point. The lab-free 'CRB-65' drops this variable.



R — RR ≥ 30 button

WHAT YOU SEE

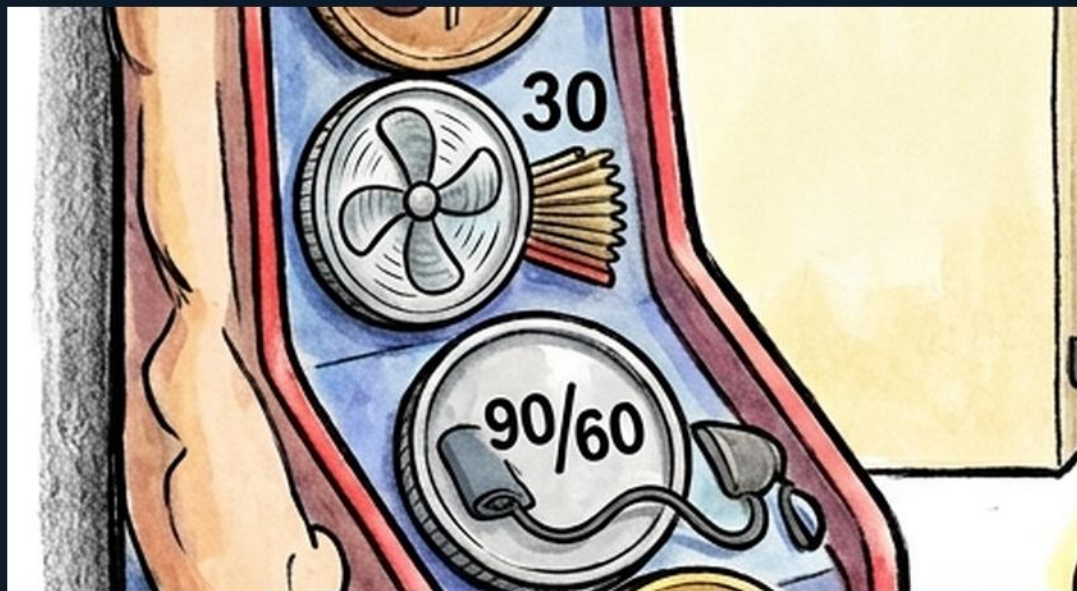
Third button: a fan/turbine for respiratory rate marked '30'.

WHAT IT ENCODES

Respiratory rate ≥ 30 breaths/min scores 1 point — an early, sensitive marker of respiratory compromise; tachypnea often precedes hypoxemia and hypotension. RR ≥ 30 is also an IDSA/ATS minor ICU criterion.

THE BOARD TRAP

"vitals look okay so skip the RR" -> wrong; tachypnea is frequently undercounted at the bedside — don't overlook it.



B — BP 90/60 button

WHAT YOU SEE

Fourth button: a BP cuff reading '90/60' — hypotension.

WHAT IT ENCODES

Hypotension (SBP <90 OR DBP ≤60) scores 1 point, signaling impending septic shock. Fluid-requiring hypotension is a minor ICU criterion; vasopressor-requiring septic shock is a MAJOR criterion (automatic ICU).

THE BOARD TRAP

"routine ward care" for a hypotensive pneumonia patient -> wrong; distinguish fluid-responsive hypotension (minor) from vasopressor-dependent shock (major -> automatic ICU).



65 — age button

WHAT YOU SEE

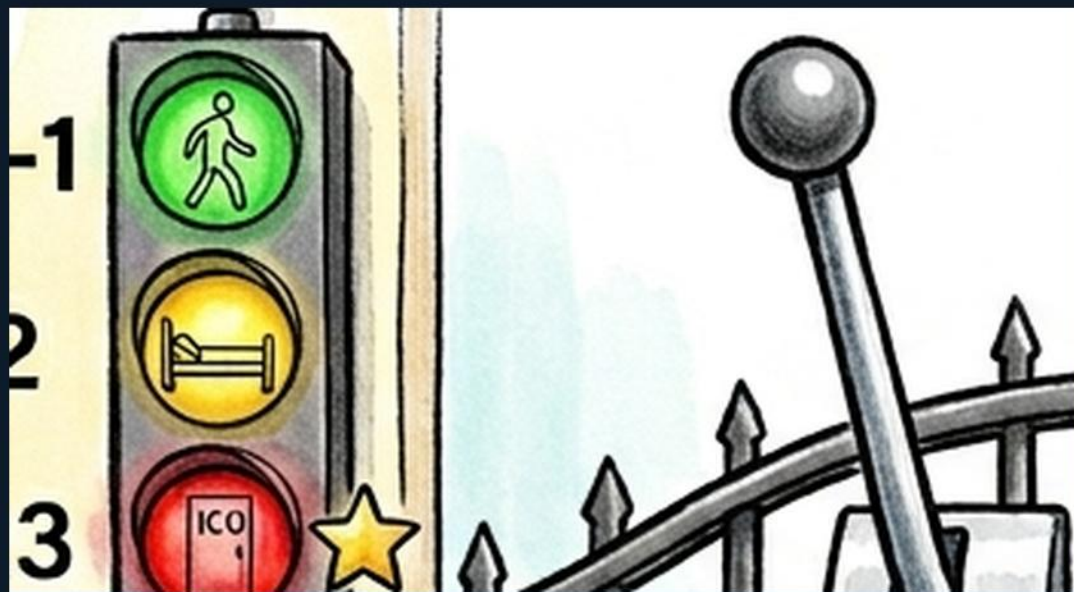
Bottom button: a birthday cake '65' — age ≥ 65 .

WHAT IT ENCODES

Age ≥ 65 scores 1 point, reflecting the steep rise in pneumonia mortality with age from comorbidity, immunosenescence, and atypical presentations.

THE BOARD TRAP

"vitals mild so send home an elder" -> wrong; age alone can tip toward admission, and elders may lack fever/leukocytosis, presenting with confusion or functional decline.



Traffic-light score (0-1 / 2 / ≥ 3)

WHAT YOU SEE

Traffic light: green 0-1, yellow 2, red star ≥ 3 — score-to-disposition.

WHAT IT ENCODES

Mortality climbs with score: 0-1 ~0.6-1.5% (outpatient), 2 = consider admission, ≥ 3 ~22% 30-day mortality (admit and assess for ICU). PSI/PORT is a more granular alternative that better identifies low-risk outpatients.

THE BOARD TRAP

"treat a CURB-65 ≥ 3 patient as outpatient" -> wrong; that is clearly unsafe — and clinical judgment/hypoxemia can override a deceptively low score.



ICU gate — major levers

WHAT YOU SEE

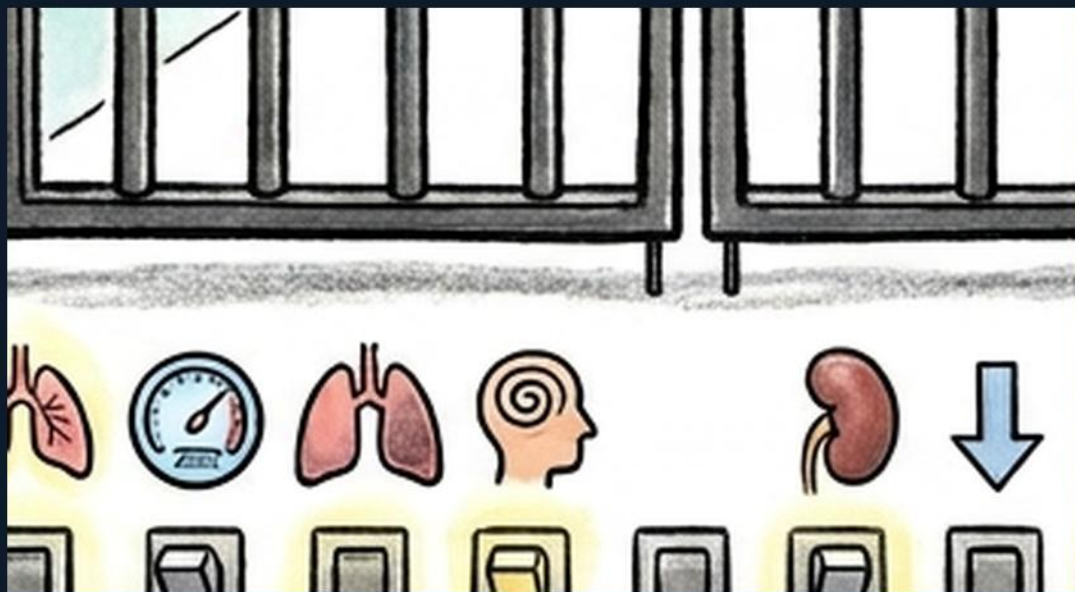
Two big levers on the ICU gate — an intubation tube and a syringe: the two major criteria.

WHAT IT ENCODES

The IDSA/ATS severe-CAP definition has 2 MAJOR criteria — invasive mechanical ventilation (respiratory failure needing intubation) and septic shock requiring vasopressors. Either ONE alone mandates ICU admission.

THE BOARD TRAP

"wait to accumulate minor criteria" -> wrong; a single major criterion is sufficient for ICU — never downgrade a ventilated or vasopressor-dependent patient to the ward.



ICU minor criteria switches (need ≥ 3)

WHAT YOU SEE

Rows of switches/icons below the gate (lungs, '250' gauge, kidneys, thermometer) — the 9 minor criteria.

WHAT IT ENCODES

Nine MINOR criteria: RR ≥ 30 , PaO₂/FiO₂ ≤ 250 , multilobar infiltrates, confusion, uremia (BUN ≥ 20), leukopenia (WBC < 4000), thrombocytopenia ($< 100k$), hypothermia ($< 36C$), and hypotension requiring aggressive fluids. Meeting ≥ 3 defines severe CAP warranting ICU/high-level monitoring.

THE BOARD TRAP

"no shock or vent so it's not severe" -> wrong; ≥ 3 minor criteria need ICU care. Note the counterintuitive ones: LEUKOPENIA (not leukocytosis), HYPOTHERMIA, and thrombocytopenia all count.



Sputum & blood cultures (bench)

WHAT YOU SEE

Magnifier over a petri dish in a hand — sputum Gram stain/culture and blood cultures before antibiotics.

WHAT IT ENCODES

Sputum Gram stain/culture (good specimen: >25 neutrophils, <10 epithelial cells/LPF) and two sets of blood cultures (before antibiotics) are for severe disease and risk factors (prior MRSA/Pseudomonas, cavitary, asplenia, leukopenia). Pneumococcus is the most common blood-culture isolate. NOT routine in mild outpatient CAP.

THE BOARD TRAP

"delay antibiotics for culture results" in severe CAP/sepsis -> wrong; collect specimens but treat promptly — time-to-antibiotics matters, and an epithelial-rich specimen is uninterpretable contamination.



Legionella & pneumococcal urinary antigen

WHAT YOU SEE

Urine cup with two gold-starred strips labeled 'Legion' and 'Pneumo' — rapid urinary antigen tests.

WHAT IT ENCODES

Urinary antigens give rapid, antibiotic-insensitive results: Legionella antigen detects *L. pneumophila* serogroup 1; pneumococcal antigen detects *S. pneumoniae*. Recommended in severe CAP and with Legionella outbreak/travel, allowing targeted de-escalation.

THE BOARD TRAP

"negative Legionella antigen excludes it" -> wrong; it misses non-serogroup-1 strains — pursue BCYE culture or PCR if suspicion is high; antigen can also stay positive for weeks (recent vs current).



Procalcitonin (crossed out)

WHAT YOU SEE

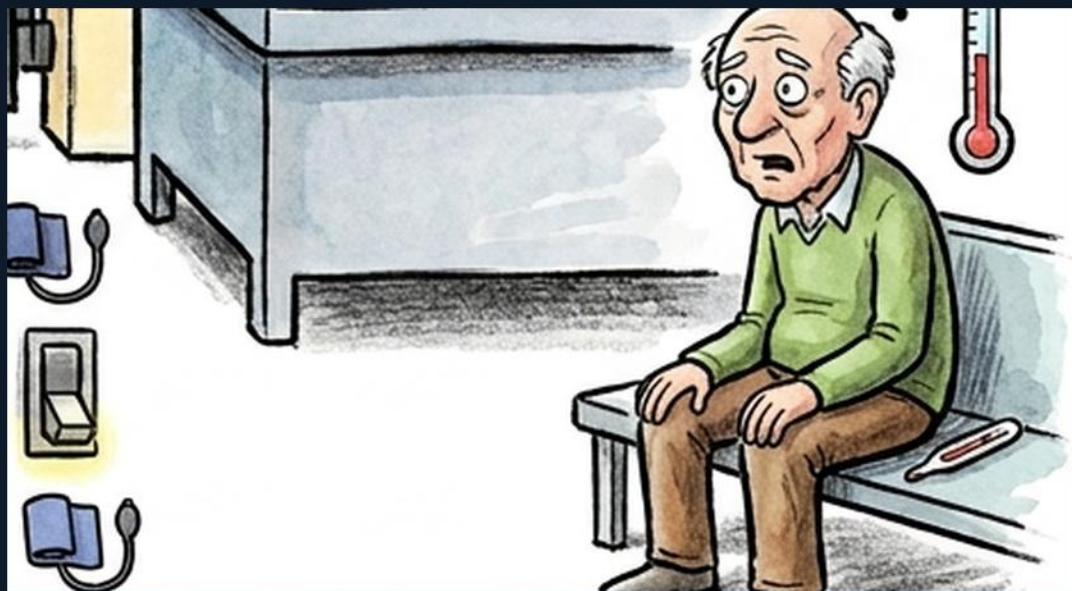
Red circle-slash over 'PCT' — do not use procalcitonin alone to START antibiotics.

WHAT IT ENCODES

Procalcitonin rises with bacterial and falls with viral/resolving infection and can help GUIDE de-escalation/duration. But IDSA/ATS say PCT should NOT be used alone to decide whether to START antibiotics in suspected CAP — clinical and radiographic diagnosis drives initiation.

THE BOARD TRAP

"low PCT, so withhold antibiotics" -> wrong; a low procalcitonin does not rule out bacterial pneumonia in a patient with a clinical/radiographic CAP diagnosis.



Elderly presentation pearl

WHAT YOU SEE

Confused old man on a bench with a thermometer and '?' — pneumonia may show only as altered mental status.

WHAT IT ENCODES

In older adults pneumonia may present WITHOUT fever, cough, or leukocytosis — altered mental status, functional decline, falls, tachypnea, or decompensation of chronic illness may be the only signs. Keep a low threshold for chest imaging.

THE BOARD TRAP

"anchor on UTI/delirium" in a confused elder -> wrong; include pneumonia for any new confusion, and don't be reassured by absent fever or normal WBC (leukopenia/hypothermia are severity markers).

Summary — every symbol

1. CURB-65 hand of 5 buttons

CURB-65 is a 5-point mortality/disposition tool, 1 point each for Confusion, Urea >7 mmol/L (BUN >19 mg/dL), Respiratory rate ≥ 30 , low Blood pressure (SBP <90 or DBP ≤ 60), and age ≥ 65

3. U — Urea >7 button

Blood urea >7 mmol/L (BUN >19 -20 mg/dL) scores 1 point, marking end-organ hypoperfusion/dehydration and worse prognosis — one reason a metabolic panel is in the initial CAP workup.

5. B — BP 90/60 button

Hypotension (SBP <90 OR DBP ≤ 60) scores 1 point, signaling impending septic shock

7. Traffic-light score (0-1 / 2 / ≥ 3)

Mortality climbs with score: 0-1 ~0.6-1.5% (outpatient), 2 = consider admission, ≥ 3 ~22% 30-day mortality (admit and assess for ICU)

9. ICU minor criteria switches (need ≥ 3)

Nine MINOR criteria: RR ≥ 30 , PaO₂/FIO₂ ≤ 250 , multilobar infiltrates, confusion, uremia (BUN ≥ 20), leukopenia (WBC <4000), thrombocytopenia ($<100k$), hypothermia ($<36C$), and hypotension requiring aggressive fluids

11. Legionella & pneumococcal urinary antigen

Urinary antigens give rapid, antibiotic-insensitive results: Legionella antigen detects L

13. Elderly presentation pearl

In older adults pneumonia may present WITHOUT fever, cough, or leukocytosis — altered mental status, functional decline, falls, tachypnea, or decompensation of chronic illness may be the only signs

2. C — Confusion button

New-onset confusion/disorientation scores 1 point, reflecting severe disease and/or hypoperfusion

4. R — RR ≥ 30 button

Respiratory rate ≥ 30 breaths/min scores 1 point — an early, sensitive marker of respiratory compromise; tachypnea often precedes hypoxemia and hypotension

6. 65 — age button

Age ≥ 65 scores 1 point, reflecting the steep rise in pneumonia mortality with age from comorbidity, immunosenescence, and atypical presentations.

8. ICU gate — major levers

The IDSA/ATS severe-CAP definition has 2 MAJOR criteria — invasive mechanical ventilation (respiratory failure needing intubation) and septic shock requiring vasopressors

10. Sputum & blood cultures (bench)

Sputum Gram stain/culture (good specimen: >25 neutrophils, <10 epithelial cells/LPF) and two sets of blood cultures (before antibiotics) are for severe disease and risk factors (prior MRSA/Pseudomonas, cavity, asplenia, leukopenia)

12. Procalcitonin (crossed out)

Procalcitonin rises with bacterial and falls with viral/resolving infection and can help GUIDE de-escalation/duration